Insert $\boldsymbol{s}(k)=\boldsymbol{s}^*(k)+\tilde{\boldsymbol{s}}(k)$ and $\boldsymbol{w}(k)=\boldsymbol{w}^{\parallel}(k)+\boldsymbol{w}^{\perp}(k)=\bar{\boldsymbol{A}}(k)K_I s^*(k)+\boldsymbol{w}^{\perp}(k)$ into Eqn. 38,

$$\begin{aligned}\bar{\boldsymbol{e}}(k+1)&=\bar{\boldsymbol{A}}(k)(I-\boldsymbol{K}_p)\bar{\boldsymbol{e}}(k)-\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i\boldsymbol{s}^*(k)-\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i\tilde{\boldsymbol{s}}(k)+\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i s^*(k)+\boldsymbol{w}^{\perp}(k)\\&=\bar{\boldsymbol{A}}(k)(I-\boldsymbol{K}_p)\bar{\boldsymbol{e}}(k)-\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i\tilde{\boldsymbol{s}}(k)+\boldsymbol{w}^{\perp}(k)\end{aligned} \tag{40}$$

We introduce the lifted state $\tilde{\zeta}_{\mathrm{PI}}(k)=\begin{bmatrix}\bar{\boldsymbol{e}}(k)\\ \tilde{\boldsymbol{s}}(k)\end{bmatrix}$ with its dynamic derived from Eqn. 39-40 as follow

$$\tilde{\zeta}_{\mathrm{PI}}(k+1)=\boldsymbol{M}_i(t)\tilde{\zeta}_{\mathrm{PI}}(k)+\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k), \tag{41}$$

where

$$\boldsymbol{M}_i(k)=\begin{bmatrix}\boldsymbol{M}_p(k) & -\boldsymbol{G}(k)\\ I & I\end{bmatrix},$$

with $\boldsymbol{M}_p(k)=\bar{\boldsymbol{A}}(p)(I-\boldsymbol{K}_p)$, $\boldsymbol{G}(k)=\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i$ and

$$\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)=\begin{bmatrix}\boldsymbol{w}^{\perp}(k)\\ -d(k)\end{bmatrix},$$

Proposition 3 (Stabilizing the PI loop reduces steady-state error) *Let $\boldsymbol{M}_p(k)=\bar{\boldsymbol{A}}(k)(\boldsymbol{I}-\boldsymbol{K}_p)$, and denote $\|\boldsymbol{K}_i\|=:h$. Assume $\sup_k\|\bar{\boldsymbol{A}}(k)\|\le M<\infty$ and $\sup_k\|\boldsymbol{M}_p(k)\|\le q<1$. If $q+Mh<1$, then the PI closed-loop control is ISS. Furthermore, the integral part exactly cancels the matched disturbance component $\boldsymbol{w}^{\parallel}$. The remaining error is due only to the unmatched component $\boldsymbol{w}^{\perp}$, which cannot be compensated. Full proof and term explanations provided in Appendix B.5*

Proof. Using the sub-multiplicativity of the induced matrix norm and the triangle inequality, and noting that $\|\boldsymbol{M}_p(k)\|\le q$, $\|\boldsymbol{G}(k)\|=\|\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i\|\le\|\bar{\boldsymbol{A}}(k)\|\|K_I\|\le Mh$, we obtain

$$\begin{aligned}\|\bar{\boldsymbol{e}}(k+1)\|&\le\|\boldsymbol{M}_p(k)\|\|\bar{\boldsymbol{e}}(k)\|+\|\boldsymbol{G}(k)\|\|\tilde{\boldsymbol{s}}(k)\|+\|\boldsymbol{w}^{\perp}(k)\|\\&\le q\|\bar{\boldsymbol{e}}(k)\|+Mh\|\tilde{\boldsymbol{s}}(k)\|+\|\boldsymbol{w}^{\perp}(k)\|,\end{aligned} \tag{42}$$

$$\|\tilde{\boldsymbol{s}}(k+1)\|\le\|\bar{\boldsymbol{e}}(k)\|+\|\tilde{\boldsymbol{s}}(k)\|+\|d(k)\|. \tag{43}$$

Introduce

$$z(k):=\begin{bmatrix}\|\bar{\boldsymbol{e}}(k)\|\\ \|\tilde{\boldsymbol{s}}(k)\|\end{bmatrix},\qquad H:=\begin{bmatrix}q & Mh\\ 1 & 1\end{bmatrix},\qquad v(k):=\begin{bmatrix}\|\boldsymbol{w}^{\perp}(k)\|\\ \|d(k)\|\end{bmatrix}.$$

Then Eqn. 42- 43 give the comparison system

$$z(k+1)\le Hz(k)+v(k). \tag{44}$$

Expanding Eqn. 44 recursively yields

$$z(k)\le H^k z(0)+\sum_{i=0}^{k-1}H^{k-1-i}v(i). \tag{45}$$

Consider the characteristic equation of H:

$$(\lambda-q)(\lambda-1)-Mh=0\iff\lambda^2-(q+1)\lambda+(q+Mh)=0.$$

Since $q+Mh<1$, the maximal root $\lambda^\star$ satisfies $\lambda^\star<1$, hence the spectral radius $\rho(H)<1$.

Let $r:=\rho(H)<1$ be the spectral radius of H. By the Gelfand formula for induced (operator) norms,

$$\lim_{k\to\infty}\|H^k\|^{1/k}=r\qquad\text{(Horn \& Johnson, 2012, p. 349).}$$

Fix any $\rho\in(r,1)$. Then, by the definition of the limit, there exists $N\in\mathbb{N}$ such that

$$\|H^k\|^{1/k}\le\rho\quad\text{for all } k\ge N\implies\|H^k\|\le\rho^k\quad\forall k\ge N.$$

Define the constant

$$C:=\max\Big\{1,\max_{0\le k\le N}\|H^k\|\rho^{-k}\Big\}.$$

Then:

- If $k \geq N$, we have $\|H^k\|\rho^{-k} \leq 1 \leq C$, hence $\|H^k\| \leq C\rho^k$.
- If $0 \leq k \leq N$, we have $\|H^k\|\rho^{-k} \leq \max_{0\leq k\leq N}\|H^k\|\rho^{-k} \leq C$, hence $\|H^k\| \leq C\rho^k$.

Therefore,

$$\|H^k\| \leq C\rho^k \qquad \text{for all } k \geq 0. \tag{46}$$

Applying Eqn. 46 to Eqn. 45 gives

$$\begin{aligned}\|z(k)\| &\leq \|H^k\|\|z(0)\| + \sum_{i=0}^{k-1}\|H^{k-1-i}\|\|v(i)\| \\ &\leq C\rho^k\|z(0)\| + C\sum_{i=0}^{k-1}\rho^{k-1-i}\|v(i)\| \\ &\leq C\rho^k\|z(0)\| + \frac{C}{1-\rho}\|v\|_\infty,\end{aligned} \tag{47}$$

where $\|v\|_\infty := \sup_{i\geq 0}\|v(i)\|$.

By construction,

$$\|\tilde{\zeta}_{PI}(k)\| = \left\|\begin{bmatrix}\bar{\boldsymbol{e}}(k)\\ \tilde{\boldsymbol{s}}(k)\end{bmatrix}\right\| = \left(\|\bar{\boldsymbol{e}}(k)\|^2 + \|\tilde{\boldsymbol{s}}(k)\|^2\right)^{1/2} = \|z(k)\|.$$

Combining this identity with Eqn. 47, the ISS estimate follows with

$$\beta(s,k) := C\rho^k s \in \mathcal{KL}, \qquad \gamma(s) := \frac{C}{1-\rho}s \in \mathcal{K},$$

which proves that the PI closed loop Eqn. 41 is ISS. □

The integral part exactly cancels the matched disturbance component $\boldsymbol{w}^{\parallel}$. The remaining error is due only to the unmatched component $\boldsymbol{w}^{\perp}$, which cannot be compensated, and to the variation rate $d(k)$ when $\bar{\boldsymbol{A}}(k)$ and $\boldsymbol{w}(k)$ change over time. In the best scenario, if a steady state exists, i.e., $\bar{\boldsymbol{A}}(k) \to \bar{\boldsymbol{A}}$ and $\boldsymbol{w}(k) \to \boldsymbol{w}$ with $\boldsymbol{w} \in \mathrm{Im}\bar{\boldsymbol{A}}$, then $\boldsymbol{w}^{\perp} \equiv 0$, $d \equiv 0$, and thus $\bar{\boldsymbol{e}}(k) \to 0$.

Remark 2 (Convergence rate versus $\boldsymbol{K}_i$) *From proposition 3, the convergence rate of $\tilde{\zeta}_{PI}(t)$ depends on ρ: the smaller ρ, the faster the convergence. We also adopt the convention (as in the proof) that $\rho \in (r,1)$. Equivalently, we examined $r(h) = \rho(H)$ and proved that with $h = \frac{(1-q)^2}{4M}$ this quantity is minimized.*

Proof Consider the characteristic polynomial of H:

$$\lambda^2 - (q+1)\lambda + (q+Mh) = 0.$$

Its discriminant is

$$\Delta(h) = (q+1)^2 - 4(q+Mh) = (q-1)^2 - 4Mh.$$

If $\Delta(h) \geq 0$ (i.e., $0 \leq h \leq \frac{(1-q)^2}{4M}$), then

$$r(h) = \frac{q+1+\sqrt{\Delta(h)}}{2},$$

and $r(h)$ decreases as h increases.

If $\Delta(h) < 0$ (i.e., $\frac{(1-q)^2}{4M} < h < \frac{1-q}{M}$), then

$$r(h) = \sqrt{q+Mh},$$

and $r(h)$ decreases as h decreases.

Hence $r(h)$ can achieve its best (smallest) value at

$$h = \frac{(1-q)^2}{4M}, \tag{48}$$

for which the error converges to zero the fastest. □

Nevertheless, as we discuss in the next section, in some situations such a large value of h may become a practical obstacle for PI control.

B.6 OVESHOOT MECHANISM

Phenomenon. A common issue in standard PI settings is *overshooting*: the closed loop oscillates around the setpoint before settling (see Åström & Hägglund (1995a, Ch. 3, §3.3)). In our terms, the integral part accumulates past error and can push the output beyond the setpoint; subsequent sign changes of the error gradually "discharge" the integral, producing a decaying oscillation. The big overshoot is undesirable when we prefer a more stable response near zero. Below we analyze the same mechanism for our PI steering setting.

Citing an observation from Vu & Nguyen (2025): in the *absence* of steering, the cosine similarity between error vectors at different layers is consistently *positive*, i.e.,

$$\cos\angle\big(\bar{\boldsymbol{e}}(i),\bar{\boldsymbol{e}}(j)\big)>0 \quad \text{for all layers } i,j,$$

so the layerwise errors share (approximately) the same direction. Consequently, with $\{\bar{\boldsymbol{x}}^{+}(k)\}$ serving as the trajectory setpoints and $\{\bar{\boldsymbol{x}}^{-}(k)\}$ the system output, an *overshoot event* occurs when the instantaneous error reverses its initial orientation, namely when

$$\langle\bar{\boldsymbol{e}}(k),\bar{\boldsymbol{e}}(0)\rangle<0.$$

We now introduce the definitions used below.

Scalarization along a direction. Let

$$v:=\frac{\bar{\boldsymbol{e}}(0)}{\|\bar{\boldsymbol{e}}(0)\|} \qquad \text{and project onto } v: \qquad \boldsymbol{e}_v(k):=v^{\top}\bar{\boldsymbol{e}}(k), \quad \boldsymbol{s}_v(k):=v^{\top}\tilde{\boldsymbol{s}}(k).$$

From the PI loop dynamics Eqn. 41 we obtain the scalar PI pair

$$\boldsymbol{e}_v(k+1)=a(k)\boldsymbol{e}_v(k)-b(k)\boldsymbol{s}_v(k)+\boldsymbol{w}_v^{\perp}(k), \tag{49}$$

$$\boldsymbol{s}_v(k+1)=\boldsymbol{s}_v(k)+\boldsymbol{e}_v(k)-d_v(k), \tag{50}$$

with $a(k)=v^{\top}\bar{\boldsymbol{A}}(k)(I-\boldsymbol{K}_p)v=v^{\top}\boldsymbol{M}_p(k)v$, $b(k)=v^{\top}\bar{\boldsymbol{A}}(k)\boldsymbol{K}_i v=v^{\top}G(k)v$, and projected disturbances $\boldsymbol{w}_v^{\perp}(k):=v^{\top}\boldsymbol{w}^{\perp}(k)$, $d_v(k):=v^{\top}d(k)$. Empirically (and consistently with the angular-steering observation in our setup), we have $v^{\top}\bar{\boldsymbol{A}}(k)v\geq 0$ for all k; together with the gain $\boldsymbol{K}_i=hI$ with $h\geq 0$, this implies $b(k)\geq 0$.

Also, the assumption $q:=\sup_k\|\boldsymbol{M}_p(k)\|$ and $M:=\sup_k\|\bar{\boldsymbol{A}}(k)\|$ yeilds

$$a(k)\leq q<1, \qquad 0\leq b(k)\leq Mh, \tag{51}$$

Since the system Eqn. 41 is ISS, so is the system Eqn. 49-Eqn. 50. In other words, both $\boldsymbol{e}_v(k)$ and $\boldsymbol{s}_v(k)$ decay. Recall that

$$\boldsymbol{s}_v(k)=\sum_{i=0}^{k-1}\boldsymbol{e}_v(i).$$

Hence, $\boldsymbol{s}_v(k)$ can only decrease when $\boldsymbol{e}_v(k)<0$, which is precisely the moment when overshoot occurs. These overshooting and decaying phenomena are observed in empirical simulation, see Fig. 6. Below, we define the overshoot in our setting.

Definition 2 (Overshoot and its amplitude) *We say an* overshoot *occurs from time k_a to k_a+m if $\boldsymbol{e}_v(k)<0\,\forall k=k_a,k_a+1,\ldots,k_a+m-1$ and $\boldsymbol{e}_v(k)\geq 0$ for $k=k_a-1,k_a+m$. Its amplitude is defined as*

$$A_a:=\max_{kt_a\leq i\leq k_a+m-1}|\boldsymbol{e}_v(i)|\,. \tag{52}$$

In standard PID settings illustrated in Åström & Hägglund (1995a, Ch. 3, §3.3)), it is observed that the overshoot amplitude decays over time. This decay is also consistent with the ISS property of the closed loop: as both $\boldsymbol{e}_v(t)$ and $\boldsymbol{s}_v(t)$ are driven down, subsequent oscillations tend to diminish in magnitude. In our simulation (see Fig. 6), the first overshoot appear to be representative. Hence, while we are not yet able to provide a formal proof, the empirical evidence and ISS intuition justify the first overshoot which is typically the dominant one and serves as a representative indicator of oscillatory behavior.

This assumption is for Proposition 4. Suppose that $a(t)\geq 0$; equivalently, $p\in(1-\frac{1}{M},1]$, which is the result of proposition 1. This assumption is expected to entail no loss of generality relative to the $|a(t)|\leq q<1$ assumption.